

RAMCO INSTITUTE OF TECHNOLOGY
Department of Mechanical Engineering
Academic Year 2019 – 2020 (Odd Semester)

Innovative practice(s) description

Course code and title: OME551 Energy Conservation and Management

Name of the Instructor(s): Mr.R.Arun Kumar, AP//Mechanical

Date & Time: 24.09.2019 & 9:50AM to 10:15AM

Use of simulation tool for environmental impact of greenhouse gases

The simulation explains how the photons from solar radiation are converted into infrared photons along with the impact in rising of environmental temperature. The inbuilt greenhouse concentration comparison between Ice Age, mid of 18th century and present will help me in making the students understand the impact of greenhouse gases on global warming. When we say orally that the global temperature had risen by 1.5°C to 2 °C in the past 100 years, it doesn't create much impact among students. But if we visually show them the rise in temperature and the reason through this simulation, they can realize the seriousness of global warming.

The simulation also shows, how the infrared photons are absorbed by atmospheric gases. This will enable us to make the students understand how traces of carbon atoms and water vapor molecules influence global warming when compared with largely available nitrogen and oxygen

OUTCOME:

Through this activity the students are made to understand the possible ways of conserving energy as mentioned above with an practical exposure which have given them an insight knowledge.

This activity helps in attaining the following CO – PO mapping:

Course Outcome / Programme Outcome	PO1	PO2	PO3	PO4	PO5	PO8	PO9	PO10	PO11	PO12
CO1 – Student will be able to explain energy auditing methodology	2	1	1	1	2	1	2	2	2	1

